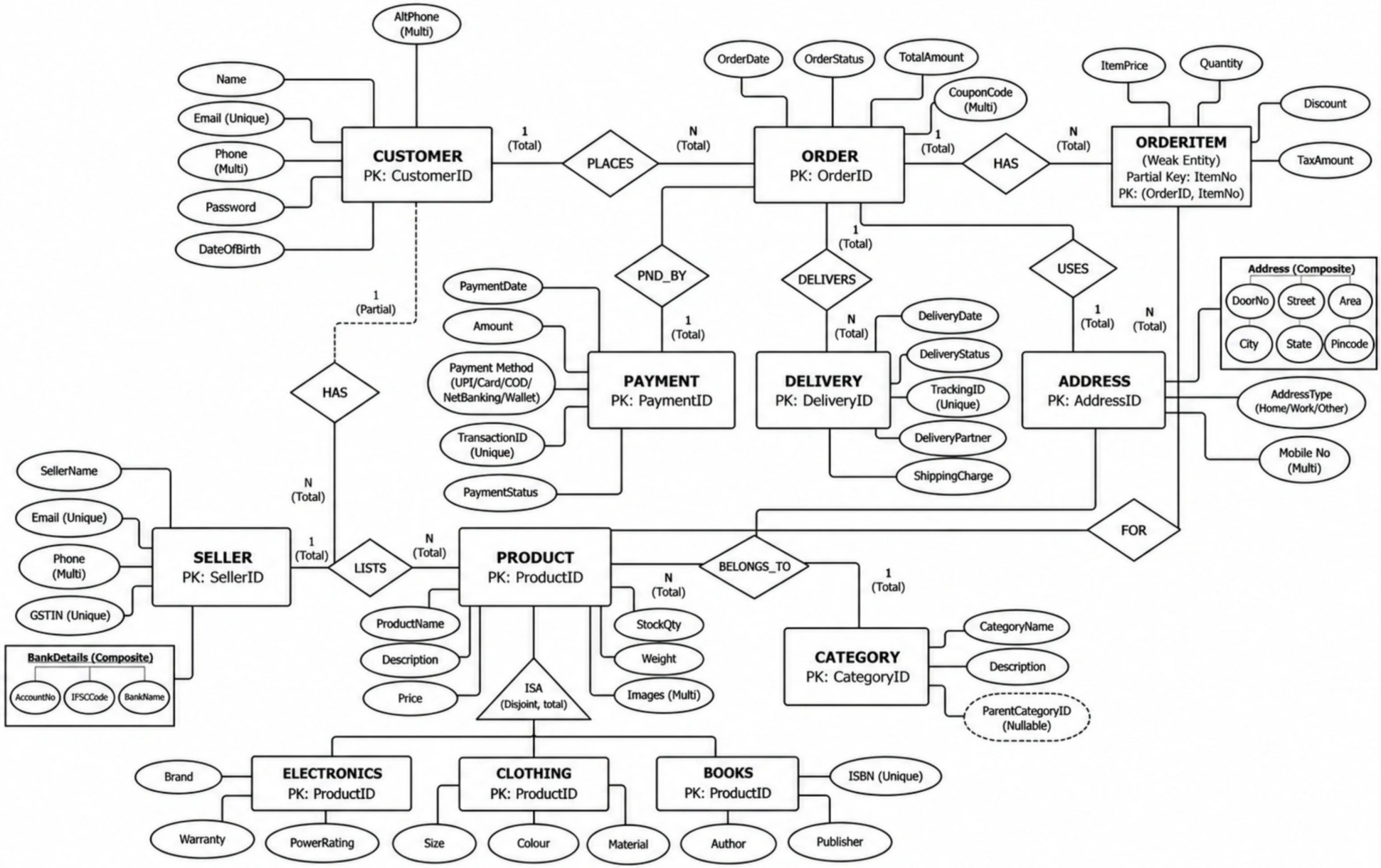
# Experiment 1

**Aim:-** Design an ER diagram for an Indian e-commerce platform with entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, Address. Specify primary keys, composite and multi-valued attributes, weak entities, specialization of Product, and participation constraints.


AltPhone (Multi)
Name
Email (Unique)
Phone (Multi)
Password
DateOfBirth
CUSTOMER
PK: CustomerID
1 (Total)
PLACES
N (Total)
OrderDate
OrderStatus
TotalAmount
CouponCode (Multi)
ORDER
PK: OrderID
1 (Total)
HAS
N (Total)
ItemPrice
Quantity
Discount
TaxAmount
ORDERITEM
(Weak Entity)
Partial Key: ItemNo
PK: (OrderID, ItemNo)
1 (Total)
PND_BY
1 (Total)
DELIVERS
N (Total)
USES
1 (Total)
N (Total)
Address (Composite)
DoorNo
Street
Area
City
State
Pincode
PaymentDate
Amount
Payment Method (UPI/Card/COD/ NetBanking/Wallet)
TransactionID (Unique)
PaymentStatus
PAYMENT
PK: PaymentID
DeliveryDate
DeliveryStatus
TrackingID (Unique)
DeliveryPartner
ShippingCharge
DELIVERY
PK: DeliveryID
ADDRESS
PK: AddressID
AddressType (Home/Work/Other)
Mobile No (Multi)
1 (Partial)
HAS
N (Total)
FOR
SellerName
Email (Unique)
Phone (Multi)
GSTIN (Unique)
SELLER
PK: SellerID
1 (Total)
LISTS
N (Total)
PRODUCT
PK: ProductID
BELONGS_TO
N (Total)
1 (Total)
BankDetails (Composite)
AccountNo
IFSCCode
BankName
ProductName
Description
Price
ISA
(Disjoint, total)
StockQty
Weight
Images (Multi)
CATEGORY
PK: CategoryID
CategoryName
Description
ParentCategoryID (Nullable)
Brand
ELECTRONICS
PK: ProductID
Warranty
PowerRating
CLOTHING
PK: ProductID
Size
Colour
Material
BOOKS
PK: ProductID
ISBN (Unique)
Author
Publisher